Biot-Savart & Ampere's Law ~London E.

Sources (MLA Format)

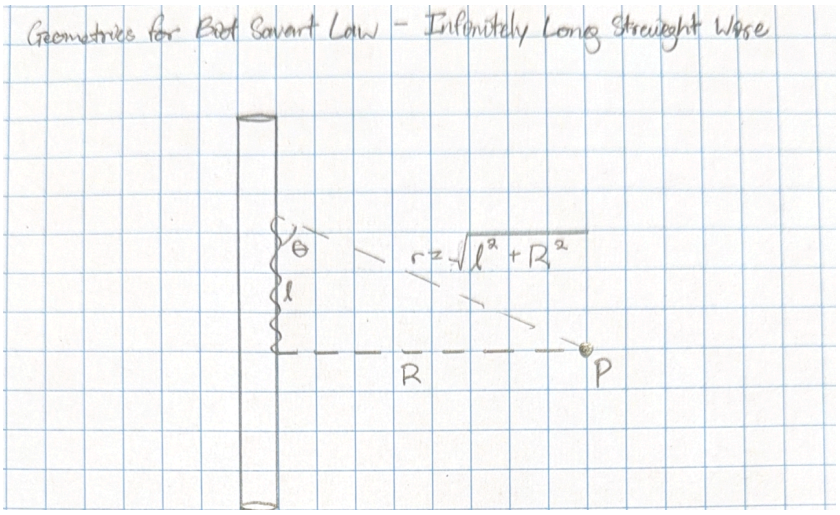
1. “12.2: The Biot-Savart Law.” *Physics LibreTexts*, Nov. 2016, phys.libretexts.org/Bookshelves/University*Physics/University_Physics*(OpenStax)/University*Physics_II-Thermodynamics_Electricity_and_Magnetism*(OpenStax)/12%3A_Sources_of_Magnetic_Fields/12.02%3A_The_Biot-Savart_Law.

Biot Savart Law

We need to show that the Biot-Savart Law can quantitatively describe a magnetic field produced by a tiny segment of any current carrying geometry. We are going to use the example of an infinitely long straight wire to demonstrate that the Biot-Savart Law can accurately describe magnetic fields quantitatively.

Variables	Meaning
l	length of a region in the geometry
B	Magnetic field
I	Current
μ	permeability of free space

Situation (Visual)



Derivation

Fundamental Equation (Biot-Savart Law)

$$d\vec{B} = \frac{\frac{\mu_0}{4\pi} \times (I(d\vec{l} \times \hat{r})}{r^2}$$

or →

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{\sin\theta d\vec{l}}{r^2}$$

Note: Using the identity

$$1 + \tan^2\theta = \sec^2\theta$$

and establishing *r* based on the situation's geometry

$$r = \sqrt{l^2 + R^2}$$

and finally establishing the value of *sine of theta* based on the situation's geometry

$$\sin\theta = \frac{R}{\sqrt{l^2 + R^2}}$$

We can then integrate to find the equation for a infinitely long straight wire using Biot-Savart's Law

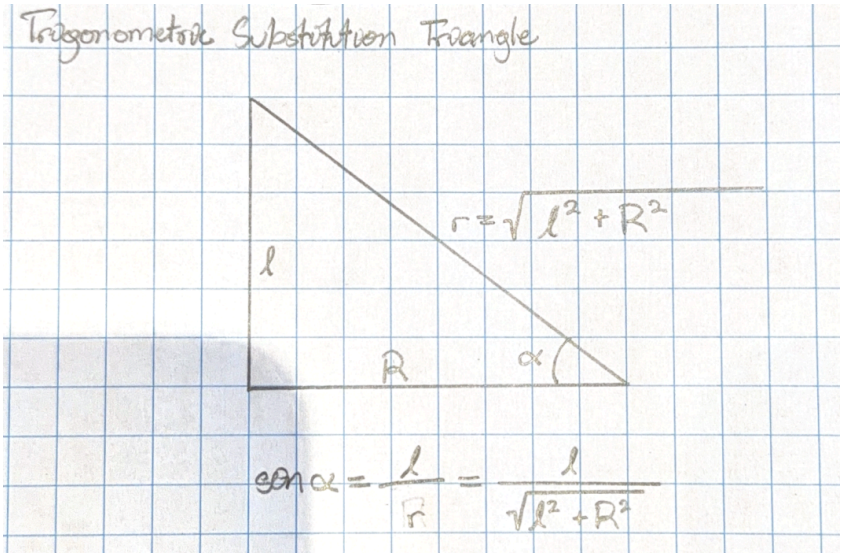
1. First we will plug in all the values established above into the integral equation

$$B = \frac{\mu_0 I}{4\pi} \int \frac{\sin\theta d\vec{l}}{(l^2+R^2)^{\frac{1}{2}}} \rightarrow \frac{\mu_0 I}{4\pi} \int \frac{\sin\theta d\vec{l}}{l^2+R^2}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{\sin\theta}{1} \frac{d\vec{l}}{(l^2+R^2)} \rightarrow \frac{\mu_0 I}{4\pi} \int \frac{R}{\sqrt{l^2+R^2}} \frac{d\vec{l}}{(l^2+R^2)}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{R d\vec{l}}{(l^2+R^2)^{\frac{3}{2}}}$$

2. Once we have plugged these values into the integral, we can now approach the trigonometric substitution. We will have to “create” a new triangle to visualize the expressions that will need to be substituted into the equation to evaluate the integral.



Now that we have our triangle, we can put this visualization on paper:

$$l = R\tan\theta$$
$$\rightarrow d\vec{l} = R\sec^2\theta$$
$$\rightarrow l^2 + R^2 = R^2\tan^2\theta = R^2(\tan^2\theta) = R^2\sec^2\theta$$
$$\rightarrow (R^2\sec^2\theta)^{\frac{3}{2}} = R^3\sec^3\theta$$

$$\rightarrow \sin\alpha = \frac{l}{\sqrt{l^2+R^2}}$$

3. Now we have our expressions to substitute in the integral. Before integrating, we need to simplify the expression:

$$R\int \frac{R\sec^2\theta d\theta}{R^3\sec^3\theta} = R\int \frac{d\theta}{R^2\sec\theta}$$
$$\frac{1}{R}\int \frac{d\theta}{\sec\theta} = \frac{1}{R}\int \cos\theta d\theta$$

4. The integral expression has been simplified, now we can integrate to get a result of:

$$\frac{1}{R}\sin\alpha + C$$

Since we did trigonometric substitution, we temporarily ignored the bounds.

5. Let us bring everything together by substituting our *sin alpha* value mentioned above into the equation, combining our constant values, and reintroducing our bounds (-∞, ∞):

$$\vec{B} = \frac{\mu_0 I}{4\pi}\left[\frac{1}{R}\sin\alpha\right] = \frac{\mu_0 I}{4\pi R}\left[\frac{l}{\sqrt{l^2+R^2}}\right]$$

$$\vec{B} = \frac{\mu_0 I}{4\pi R}\left[\frac{\infty}{\sqrt{\infty^2+R^2}} - \left(-\frac{\infty}{\sqrt{(-\infty)^2+R^2}}\right)\right] = \frac{\mu_0 I}{4\pi R}[1-(-1)]$$

6. Our final result will be the following:

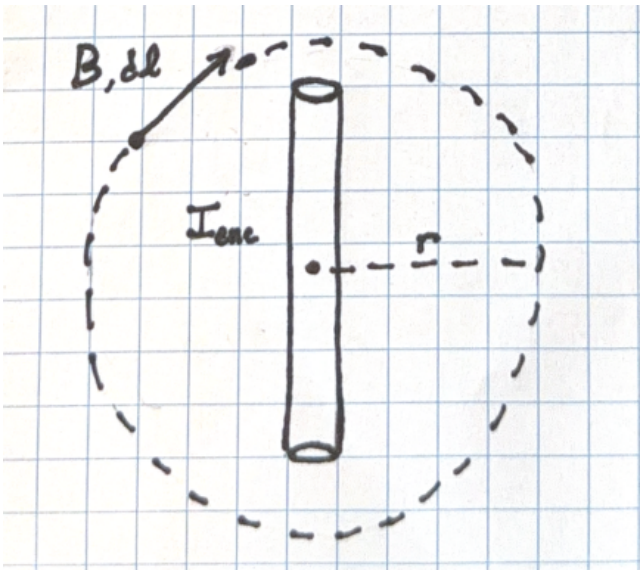
$$\boxed{\frac{\mu_0 I}{2\pi R}}$$

Ampere’s Law

Ampere’s Law relates a line integral of B around a closed loop to the current passing through the surface bounded by that loop. This Law significantly reduces the amount of derivation needed to quantitatively describe the magnetic field produced by a tiny segment of any current carrying geometry. The same example will be used: An infinitely long, straight current carrying wire.

Variables	Meaning	Units
□ B□ · dl□	Closed line integral of B along the Amperian Loop	T*m
μ_0	Permeability of free space	T* (m/A)
I_enc	Net current threading the surface bounded by the loop (sign from RHR)	A
dl□	Infinitesimal length element along the closed path (vector, tangent to path)	m

Situation (Visual)



Derivation

We need to keep in mind that the Magnetic Field is constant everywhere and that Ampere's Law is reliant on symmetry.

1. We start off with Ampere's Law:

$$\oint \vec{B} \cdot d\vec{I} = \mu_0 I_{enc}$$

   or →

$$\oint \vec{B} \cdot d\vec{I} \cos\theta = \mu_0 I_{enc}$$

2. Since B and *dl* are parallel everywhere, *theta* is equal to zero, thus making *cosine of theta* equal to one, giving us the following result:

$$\oint Bdl(1) = \mu_0 I_{enc}$$

3. Since our magnetic field is constant, B is a constant and can therefore be brought to the outside of the integral:

$$B \oint dl = \mu_0 I_{enc}$$

4. We are working with a loop (circle). The integral of *dl* in this case would be equal to the circumference of a circle:

$$B(2\pi r) = \mu_0 I_{enc}$$

5. Now we can solve for our magnetic field, B:

$$\frac{B(2\pi r)}{2\pi r} = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{\mu_0 I}{2\pi r}$$